

Logistic Regression (1)

(로지스틱 회귀분석 (1))

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Today's Topics

1. Logistic Regression

- Inference on the Regression Coefficient
- Hypothesis Testing about Regression Coefficient
- Distribution of Regression Coefficient
- Model Comparison
- Goodness-of-fit for Ungrouped Data and Grouped Data
- Diagnostic of Logistic Regression
- Logistic Regression for Qualitative Variables
- $2 \times 2 \times K$ Contingency Table

Logistic Regression

Definition

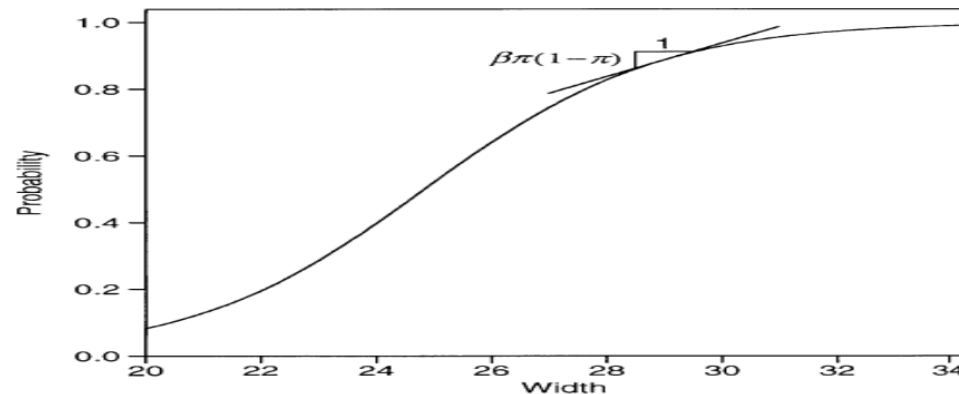
Y : binary response variable

X : quantitative explanatory variable

When $X = x$, the probability of success, $\pi(x)$, can be calculated by

$$g(\pi(x)) = \beta_0 + \beta_1 x, \quad g(x) = \log \frac{x}{1-x} \Rightarrow \pi(x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}$$

Thus $\pi(x)$ is a S-curved shaped (sigmoid) function:



Logistic Regression (Cont'd)

Meaning of regression coefficients (β_1)

Sign determines whether it is either increasing or decreasing

Amount of its absolute value determines the increase/decrease rate

If $\beta_1 = 0$, $\pi(x)$ is a constant and is not affected by x

X does not affect Y

Slope of the tangent

$$\frac{\partial \pi(x)}{\partial x} = \frac{\beta_1 e^{\beta_0 + \beta_1 x}}{(1 + e^{\beta_0 + \beta_1 x})^2} = \beta_1 \pi(x)(1 - \pi(x))$$

If $\pi(x)$ goes to 1 or 0, it becomes 0

If $\pi(x)$ is equal to 0.5, it is the steepest

Median effective level: $x = -\beta_0/\beta_1$ ($\pi(x) = 0.5$)

Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

```
> head(crabs)
```

	Satellites	Width	Dark	GoodSpine	Rep1	Rep2	HasSatellite
1	8	28.3	no	no	2	2	1
2	0	22.5	yes	no	4	5	0
3	9	26.0	no	yes	5	6	1
4	0	24.8	yes	no	6	6	0
5	4	26.0	yes	no	6	8	1
6	0	23.8	no	no	8	8	0

```
> summary(model_linear)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.76553	0.42136	-4.190	4.46e-05 ***
Width	0.09153	0.01597	5.731	4.42e-08 ***

```
> summary(model_logistic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-12.3508	2.6287	-4.698	2.62e-06 ***
Width	0.4972	0.1017	4.887	1.02e-06 ***

```
require(glm2)
```

```
data(crabs)
```

```
crabs$HasSatellite <- ifelse(crabs$Satellites > 0, 1, 0)
```

```
# Model 1
```

```
model_linear <- glm(HasSatellite ~ Width, family =  
"gaussian", data = crabs)
```

```
summary(model_linear)
```

```
# Model 2
```

```
model_logistic <- glm(HasSatellite ~ Width, family =  
binomial(link = "logit"), data = crabs)
```

```
summary(model_logistic)
```

Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

Model 1: linear regression

$$\hat{\pi}(x) = \hat{\beta}_0 + \hat{\beta}_1 x = -1.77 + 0.09x$$

If $x = 33.5$, $-1.77 + 0.09 \times 33.5 = 1.25 > 1$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.76553	0.42136	-4.190	4.46e-05	***
Width	0.09153	0.01597	5.731	4.42e-08	***

Model 2: logistic regression

$$\hat{\pi}(x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)} = \frac{\exp(-12.35 + 0.50x)}{1 + \exp(-12.35 + 0.50x)}$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.3508	2.6287	-4.698	2.62e-06	***
Width	0.4972	0.1017	4.887	1.02e-06	***

Median effective level?

If $x = 26.5$, what is the slope of the tangent?

Logistic Regression (Cont'd)

Odds ratios

Definition of odds ratios

Odds:

$$\frac{\pi(x)}{1 - \pi(x)} = e^{\beta_0 + \beta_1 x} = e^{\beta_0} e^{\beta_1 x}$$

Odds ratio:

$$\theta(x_1, x_2) = \frac{\pi(x_1)/[1 - \pi(x_1)]}{\pi(x_2)/[1 - \pi(x_2)]} = e^{\beta_1(x_1 - x_2)}$$

Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.3508	2.6287	-4.698	2.62e-06	***
Width	0.4972	0.1017	4.887	1.02e-06	***

Interpretation

$$\hat{\beta}_1 = 0.50$$

If the width becomes 1cm bigger, the odds increases by $e^{0.5} = 1.64$

Inference on the Regression Coefficient

Confidence interval of regression coefficient

Wald confidence interval - based on the normality of MLE

Confidence interval of β

$$(L, U) \equiv \hat{\beta} \pm Z_{1-\frac{\alpha}{2}} \times \hat{\sigma}(\hat{\beta})$$

Confidence of odds ratio on the unit increase

$$(e^L, e^U), L = \hat{\beta} - Z_{1-\frac{\alpha}{2}} \times \hat{\sigma}(\hat{\beta}), U = \hat{\beta} + Z_{1-\frac{\alpha}{2}} \times \hat{\sigma}(\hat{\beta})$$

Inference on the Regression Coefficient (Cont'd)

Confidence interval of regression coefficient

Profile confidence interval - based on the likelihood function

Confidence interval derived by likelihood function

Let $l_P(\theta_1) = \max_{\theta_2} \log L(\theta_1, \theta_2) = \log L(\theta_1, \hat{\theta}_2(\theta_1))$, $l = \max_{\theta_1, \theta_2} \log L(\theta_1, \theta_2)$

$\Rightarrow$ Confidence interval of θ_1 :

$$\{\theta_1 | l(\hat{\theta}_1, \hat{\theta}_2) - l_P(\theta_1) < 0.5\chi_{1-\alpha}^2(df = 1)\}$$

$\Rightarrow$ Confidence interval of θ_2 :

$$\{\theta_2 | l(\hat{\theta}_1, \hat{\theta}_2) - l_P(\theta_2) < 0.5\chi_{1-\alpha}^2(df = 1)\}$$

Profile confidence interval is more accurate than Wald confidence interval

Inference on the Regression Coefficient (Cont'd)

Example

What is the regression coefficient of width and confidence interval of odds ratio by the unit increase?

```
> summary(model_logistic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.3508	2.6287	-4.698	2.62e-06	***
Width	0.4972	0.1017	4.887	1.02e-06	***

```
> confint(model_logistic,test="LRT")
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-17.8100090	-7.4572470
Width	0.3083806	0.7090167

```
> confint(model_logistic,test="Rao")
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-17.4684774	-7.2452731
Width	0.2995764	0.6953433

95% confidence interval: $0.497 \pm 1.96 \times 0.102 = (0.297, 0.697)$

$(e^{0.297}, e^{0.697}) = (1.35, 2.01)$

Hypothesis Testing about Regression Coefficient

Hypothesis testing

Hypothesis

$$H_0: \beta = 0 \text{ vs } H_1: \beta \neq 0$$

Wald test

$$Z = \frac{\hat{\beta}}{ASE} \sim N(0,1), \text{ or } W = \left(\frac{\hat{\beta}}{ASE}\right)^2 \sim \chi^2(df = 1)$$

Likelihood ratio test

$$-2 \log \left(\frac{L_0}{L_1} \right) = -2[\log(L_0) - \log(L_1)] \sim \chi^2(df = 1)$$

Likelihood ratio test generally performs better than Wald test

Hypothesis Testing about Regression Coefficient (Cont'd)

(Practice) Horseshoe crabs data

Wald test

```
> summary(model_logistic)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.3508	2.6287	-4.698	2.62e-06	***
Width	0.4972	0.1017	4.887	1.02e-06	***

Likelihood ratio test

```
> logLik(model_logistic_red)
```

```
'log Lik.' -112.8793 (df=1)
```

```
> logLik(model_logistic)
```

```
'log Lik.' -97.22633 (df=2)
```

```
> lmtest::lrtest(model_logistic_red, model_logistic)
```

Likelihood ratio test

Model 1: HasSatellite ~ 1

Model 2: HasSatellite ~ Width

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	1	-112.879			
2	2	-97.226	1	31.306	2.204e-08 ***

```
model_logistic_red <- glm(HasSatellite ~ 1, family  
= binomial(link = "logit"), data = crabs)
```

```
logLik(model_logistic)  
logLik(model_logistic_red)
```

```
lmtest::lrtest(model_logistic_red, model_logistic)
```

Hypothesis Testing about Regression Coefficient (Cont'd)

Hypothesis testing with a small sample

- Exact test

 - Conditional logistic model

- Resampling methods

 - Bootstrap

Distribution of Regression Coefficient

When sufficiently large samples are used,

Property of MLE

$$\hat{\beta}_0 \sim N(\beta_0, V(\hat{\beta}_0)), \quad \hat{\beta}_1 \sim N(\beta_1, V(\hat{\beta}_1))$$

$$\Rightarrow \hat{\beta}_0 + \hat{\beta}_1 x \sim N(\beta_0 + \beta_1 x, V(\hat{\beta}_0 + \hat{\beta}_1 x))$$

$$\Rightarrow V(\hat{\beta}_0 + \hat{\beta}_1 x) = V(\hat{\beta}_0) + V(\hat{\beta}_1 x) + 2x \text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$$

Denote the estimate of $\sqrt{V(\hat{\beta}_0 + \hat{\beta}_1 x)}$ by ASE

$$Z = \frac{\hat{\beta}_0 + \hat{\beta}_1 x - (\beta_0 + \beta_1 x)}{ASE} \sim N(0, 1)$$

100(1 - α)% confidence interval of $\hat{\beta}_0 + \hat{\beta}_1 x$

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm z_{1-\alpha/2} ASE = [L, U]$$

100(1 - α)% confidence interval of $\pi(x)$ $\hat{\pi}(x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)} \Rightarrow \left[\frac{e^L}{1 + e^L}, \frac{e^U}{1 + e^U} \right]$

Distribution of Regression Coefficient (Cont'd)

(Practice) Horseshoe crabs data

```
> covariance_matrix <- vcov(model_logistic)
```

```
> covariance_matrix
```

	(Intercept)	Width
(Intercept)	6.9101576	-0.26684761
Width	-0.2668476	0.01035012

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.3508	2.6287	-4.698	2.62e-06	***
Width	0.4972	0.1017	4.887	1.02e-06	***

Interpretation

If $x = 26.5$, $\hat{\beta}_0 + \hat{\beta}_1 x = -12.351 + 0.497 \times 26.5 = 0.825$

R results: $V(\hat{\beta}_0) = 6.91$, $V(\hat{\beta}_1) = 0.0104$, $Cov(\hat{\beta}_0, \hat{\beta}_1) = -0.267$

$$\Rightarrow ASE = \sqrt{V(\hat{\beta}_0 + \hat{\beta}_1 x)} = 0.2498$$

95% confidence interval of $\beta_0 + \beta_1 x$

$$0.825 \pm 1.96 \times 0.2498 = [0.335, 1.315]$$

95% confidence interval of $\pi(x)$

$$\left[\frac{e^{0.335}}{1 + e^{0.335}}, \frac{e^{1.315}}{1 + e^{1.315}} \right] = [0.583, 0.788]$$

Model Comparison

Goodness-of-fit

Evaluate the model by comparing the expected and observed frequencies

If the expected frequency and the observed frequency are similar, the model is considered as an appropriate model

If the fit is not good, each observation is evaluated using residuals and other diagnostic measures, and identify the reason for the bad fit

Pearson X^2 and/or deviance G^2 can be used

If the value of the fitness test statistic is large, X^2 and G^2 simply indicate that there is the lack of fit, but they do not provides the reason

Some clues can be identified by comparing it with more complex models

Model Comparison (Cont'd)

Pearson X^2 test

$$X^2 = \sum \frac{(y_i - \hat{\mu}_i)^2}{\text{var}(y_i)} = \sum \frac{(y_i - n_i \hat{\pi}_i)^2}{n_i \hat{\pi}_i (1 - \hat{\pi}_i)} \sim \chi^2(df)$$

df : sample size - the number of parameters

Deviance G^2 test

l_M : $\max \log L_M$

l_S : $\max \log L_S$

Scaled Deviance: $-2(l_M - l_S) \sim \chi^2(df)$

df : sample size - the number of parameters

Model Comparison (Cont'd)

(Practice) Snoring data

```
> snore
```

	snore_score	yes	no	trials	prop
1	0	24	1355	1379	0.01740392
2	2	35	603	638	0.05485893
3	4	21	192	213	0.09859155
4	5	30	224	254	0.11811024

```
> summary(snore_model_logistic)
```

Call:

```
glm(formula = cbind(yes, no) ~ snore_score, family = binomial(link = "logit"),  
     data = snore)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.86625	0.16621	-23.261	< 2e-16 ***
snore_score	0.39734	0.05001	7.945	1.94e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 65.9045 on 3 degrees of freedom
Residual deviance: 2.8089 on 2 degrees of freedom
AIC: 27.061

```
## Snoring and Heart disease
```

```
snore <- data.frame(snore_score = c(0, 2, 4, 5), yes = c(24, 35,  
21, 30), no = c(1355, 603, 192, 224))
```

```
snore$trials <- snore$yes + snore$no
```

```
snore$prop <- snore$yes / snore$trials
```

```
snore_model_logistic <- glm(cbind(yes, no) ~ snore_score,  
data=snore, family=binomial(link="logit"))
```

```
summary(snore_model_logistic)
```

```
pchisq(2.8089,df=2,lower=F); 0.2455
```

Model Comparison with Likelihood Ratio Test

Deviance

S : saturated model

For each observation/subject, a single parameter is estimated and the largest number of parameters is fit

For different value of x_i , the probability is estimated by

$$\hat{\pi}(x_i) = y_i/n_i$$

Scaled deviance

$$G^2(M) = -2\{\log(L_M) - \log(L_S)\}$$

Model Comparison with Likelihood Ratio Test (Cont'd)

Relationship between deviance and likelihood ratio test

Model definition

M_1 : reduced model (some parameters for M_2 are set to be 0) vs M_2 : full model

Likelihood ratio test

$$G^2(M_1|M_2) \equiv G^2(M_1) - G^2(M_2) = -2\{\log(L_1) - \log(L_2)\}$$

If the model fit by M_1 is worse than M_2 , likelihood ratio test statistics becomes larger

Model Comparison with Likelihood Ratio Test (Cont'd)

(Practice) Snoring data

```
> summary(snore_model_logistic_red)
```

```
Call:
glm(formula = cbind(yes, no) ~ 1, family = binomial(link = "logit"),
    data = snore)
```

Coefficients:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.07185    0.09753   -31.5   <2e-16 ***
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 65.904  on 3  degrees of freedom
Residual deviance: 65.904  on 3  degrees of freedom
AIC: 88.157
```

```
snore_model_logistic_red
<- glm(cbind(yes, no) ~ 1, data=snore,
family=binomial(link="logit"))
summary(snore_model_logistic_red)
pchisq(65.904,df=3,lower=F); 3.213e-14

pchisq(63.09,df=1,lower=F); 1.975e-15
```

Interpretation

$$G^2(M_1) = 65.90 \sim \chi^2 (df = 3), G^2(M_2) = 2.81 \sim \chi^2 (df = 2)$$

$$\text{Likelihood ratio test: } G^2(M_1|M_2) = 65.90 - 2.81 = 63.09 \sim \chi^2 (df = 1)$$

Goodness-of-fit for Ungrouped Data and Grouped Data

Grouped vs ungrouped data

Grouped data: the case where $n_i \geq 2$ for each observation

Ungrouped data: the case where $n_i = 1$

(Example) crab and snoring data are grouped data?

Goodness-of-fit for Ungrouped Data and Grouped Data (Cont'd)

Problem of ungrouped data

For ungrouped data, $n_i = 1$ and degrees of freedom for X^2 , G^2 are:

sample size - the number of parameters

⇒ Sample size and degree of freedom are similar, and X^2 , G^2 do not follow the χ^2 distribution

Goodness-of-fit for Ungrouped Data and Grouped Data (Cont'd)

Solution

For each observation, calculate the probability, $\pi(x_i)$, and use it to group each observation

Then for each group, compare the observed and expected counts

There are two different ways to make groups

- (1) Groups with the same interval of $\pi(x_i)$
- (2) Groups with the same sample sizes (popular)

(Example) Hosmer-Lemeshow test

Goodness-of-fit for Ungrouped Data and Grouped Data (Cont'd)

Hosmer-Lemeshow test

Sort observations with $\hat{\pi}$, and categorize into g groups

How to categorize samples

(1) Assign the same or similar observations for each group

The number of observation for each group = the total sample size/ g

(2) Categorize observations with the determined value with the same length of predicted probability

Goodness-of-fit for Ungrouped Data and Grouped Data (Cont'd)

Hosmer-Lemeshow test

$$\hat{C} = \sum_{k=1}^g \frac{(o_k - n_k \bar{\pi}_k)^2}{n_k \bar{\pi}_k (1 - \bar{\pi}_k)} \sim \chi^2(df = g - 2)$$

n_k : the number of subjects belong to group k

o_k : the number of subjects with success in group k

$\bar{\pi}_k$: the sample proportion of subjects with success in group k

$n_k \bar{\pi}_k$: the expected number of subjects with success in group k

Goodness-of-fit for Ungrouped Data and Grouped Data (Cont'd)

(Practice) Horseshoe crabs data

Hosmer and Lemeshow goodness of fit (GOF) test

data: actual_response, predicted_probs
X-squared = 4.3855, df = 8, p-value = 0.8208

```
# Hosmer-Lemeshow test
require(ResourceSelection)
predicted_probs <- fitted(model_logistic)
actual_response <- crabs$HasSatellite
hoslem.test(actual_response, predicted_probs, g = 10)
```

Interpretation

If $g = 10$, $\hat{C} = 4.3855$ and $P = 0.8208$ (thus, better fit)

Diagnostic of Logistic Regression

Model diagnostic

G^2 and X^2 : both are measures which quantify the quality of model fit, and useful to detect the lack of fit

However they cannot be used to identify the reason, and alternatively, Pearson and deviance residuals can be used

Pearson residual

y_i : the number of success for the observation i with n_i trials

$\hat{\pi}_i$: the estimated probability of success

$$e_i = \frac{y_i - n_i \hat{\pi}_i}{\sqrt{\text{var}(y_i)}} \sim N(0,1)$$

Diagnostic of Logistic Regression (Cont'd)

Adjusted Person residual

h_i : the leverage estimated from the diagonal element of hat matrix

$$r_{P,i} = \frac{y_i - n_i \hat{\pi}_i}{\sqrt{\text{var}(y_i - n_i \hat{\pi}_i)}} = \frac{y_i - n_i \hat{\pi}_i}{\sqrt{n_i \hat{\pi}_i (1 - \hat{\pi}_i) (1 - h_i)}} \sim N(0,1)$$

Hat matrix: $H = W^{1/2} X (X^t W X)^{-1} X^t W^{1/2}$

Deviance residual

Adjusted deviance residual: $r_{D,i} = \text{sign}(y_i - \hat{\mu}_i) \frac{\sqrt{d_i}}{\sqrt{a(\psi)(1 - h_i)}}$, $d_i = 2a(\psi)(\log L_{i,S} - \log L_{i,M})$

Likelihood residual: $r_{L,i} = \text{sign}(y_i - \mu_i) \sqrt{(1 - h_i)r_{D,i}^2 + h_i r_{P,i}^2}$

Diagnostic of Logistic Regression (Cont'd)

(Practice) Snoring data

	Pearson	Adj_Pearson	Deviance	Adj_Deviance
1	-0.8131634	-1.6783847	-0.8346344	-1.722701
2	1.2968557	1.5448873	1.2520731	1.491540
3	0.2781891	0.3225535	0.2757747	0.319754
4	-0.6736948	-1.1970179	-0.6845134	-1.216240

```
# Diagnostic of Logistic Regression
r1 <- residuals(snore_model_logistic, type="pearson")
r2 <- rstandard(snore_model_logistic, type = "pearson")
r3 <- residuals(snore_model_logistic, type="deviance")
r4 <- rstandard(snore_model_logistic, type = "deviance")

resi <-
data.frame(Pearson=r1,Adj_Pearson=r2,Deviance=r3,Adj
_Deviance=r4); resi
```